Unified Network Embedding via Mutual Fusion of Communities and Roles

Pengfei Jiao, *Member, IEEE*, Wang Zhang, Xuan Guo, Huan Liu, Yanxian Bi, Yefei Zhang, and Zhidong Zhao

***Abstract*—Most network embedding (NE) methods are either based on the proximity for community-guided tasks or on the structural similarity for role-oriented tasks. While being prevalent and effective, there still exists some potential issues that need further attention: 1) community and role are always regarded as orthogonal problems. They have rarely been combined to model the latent structures within the complex network. However, the generation of the network is usually jointly driven by these two mechanisms; and 2) few works study the interaction between roles or communities, which leads to the generation process of the network cannot being effectively modeled. To solve these problems, we propose a unified network embedding framework via mutual fusion of community and role (UMFCR). We combine the Gaussian mixture model (GMM) with a variational graph auto-encoder to generate node embeddings and discover the membership distribution of each node. An elaborate fusion pattern is then designed to produce the generation process for each link from the perspective of both community and role. The promising experimental results on real-world data demonstrate the necessity of fusing these two mechanisms and the superior performance of the model on different network tasks.**

***Index Terms*—Community detection, generative model, network embedding (NE), role discovery.**

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Pengfei Jiao and Zhidong Zhao are with the School of Cyberspace, Hangzhou Dianzi University, Hangzhou 310018, China, and also with Zhejiang Provincial Key Laboratory for Sensitive Data Security Protection and Confidentiality Management, Hangzhou Dianzi University, Hangzhou 310018, China (e-mail: pjiao@hdu.edu.cn; zhaozd@hdu.edu.cn).

Wang Zhang and Xuan Guo are with the College of Intelligence and Computing, Tianjin University, Tianjin 300350, China (e-mail: wangzhang@tju.edu.cn; guoxuan@tju.edu.cn).

Huan Liu and Yefei Zhang are with the School of Cyberspace, Hangzhou Dianzi University, Hangzhou 310018, China (e-mail: huanliu@hdu.edu.cn; zhangyf_valora@hdu.edu.cn).

Yanxian Bi is with China Academy of Electronic and Information Technology, CETC Academy of Electronics and Information Technology Group Company Ltd., Beijing 100041, China (e-mail: biyanxian@cetc.com.cn).

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I. Introduction

COMMUNITY and role are two fundamentally different but complementary concepts [1], [2], [3]. The community identifies sets of nodes with a higher inner connection density than outer, and the role determines such sets of nodes that the functions inside the set are more similar than the outside. Community structure and structural role exist in a wide variety of real-world systems, ranging from social networks [4], biological protein-protein networks [5], air-traffic networks [6], [7], citation networks [8], [9], etc. They correspond to specific semantics in different scenarios. For example, in social networks, communities may represent groups with common interests, and roles represent groups with the same positions. In citation networks, communities indicate sets with the same research field, and roles indicate sets with similar influence. Intuitively, community detection and role discovery provide orthogonal perspectives for network analysis, allowing us to discover the potential interactions [5], abnormal behavior patterns [10], and even evolutionary trends [11]. They play an important role in understanding the formation and evolution of networks.

Network embedding (NE) [12], [13], [14], [15], [16], [17] has achieved the encouraging performance with powerful expressive ability over the past decades, especially in community detection and role discovery [1], [2]. For communities, the methods [e.g., DeepWalk [18], GraphSTONE [19], and SDNE [20] etc.] encode the proximity of nodes such that nodes with close spatial distances have similar low-dimensional representations. For role, the methods [e.g., RolX [21], RESD [22], and GAS [23], etc.] are concerned with the expression of structural similarity so that nodes with similar functions can be embedded in closer positions. In addition, there are also some probabilistic generative frameworks for modeling complex networks, such as VGAE [24] and DGLFRM [25]. While being prevalent and effective, these methods still exist some potential problems that require further attention. First, community and role are traditionally regarded as orthogonal problems and they have rarely been combined to model the latent structures within complex networks. In community-oriented methods, they usually neglect to capture the interactions between distant nodes which is very important in role discovery task. This phenomenon is particularly evident in methods based on traditional random walk [18], which only construct the context sequence with the neighbor nodes of the target node. On the contrary, role-oriented methods focus too much on the connection patterns of nodes, while ignoring the similarities between adjacent nodes. For example, in matrix factorization-based methods [21], [26], they usually only apply various matrix factorization techniques to the structural similarity matrix of nodes. However, the generation of complex networks is usually jointly driven by these two mechanisms and is difficult to distinguish. A single-side description is not sufficient to make it contain enough information to support downstream tasks. Furthermore, the vast majority of methods are based on proximity or structural similarity, and few works study the interactions between roles or communities, which leads to the generation of the network cannot be effectively modeled. Intuitively, the connection between each pair of nodes in any scenario is strongly associated with their communities or roles. For example, members within the same club (community) naturally have more connections, while each club usually has a specific interaction mechanism (formed by their roles). Considering the links between different clubs, they are related to the semantic information of the community and the role identities of the nodes, e.g., the basketball club is more likely to be active with the football club than the reading club, and this activity is usually initiated by the bridge nodes. Among the few related methods based on generative frameworks [27], they assign groups to nodes and then establish edges according to the probability of interactions between different groups. However, such methods tend to capture the community structure rather than the structural roles.

To the best of our knowledge, Node2vec [28], SPINE [29], REACT [30], and SPaE [31] are the most relevant to the above problems, which try to simultaneously capture proximity and structural similarity of nodes. Node2vec and SPINE add flexibility in visiting neighborhoods by defining a biased random walk strategy, to control whether the learned node representation is more toward the community or the role. However, they usually require manual adjustment of hyperparameters to achieve this goal. REACT and SPaE, are methods based on matrix factorization, the former defines a specific regularization term to capture the diversity relationship between roles and communities, while the latter defines the nonnegative parameter that balances the contribution of structural proximity and equivalence on the embedding results. However, they have high complexity and are difficult to be applied to large-scale networks. In addition, the methods mentioned above also do not consider the interactions between different groups (i.e., communities and roles).

In this work, we propose a unified network embedding framework via mutual fusion of community and role (UMFCR) to address the above problems. The basic assumption behind our method is that the interactions between nodes are related to both community and role memberships. Specifically, we combine the Gaussian mixture model (GMM) with VGAE to generate the node embeddings and discover the membership distributions from two perspectives of community and role, respectively. Using the mixture-of-Gaussians prior instead of the single Gaussian prior can effectively model more complex data distributions. Furthermore, we design the elaborate fusion pattern, which flexibly portrays the role interactions within and between communities, so that we can consider both the influence of community and role on link generation at the same time. Our contributions can be summarized as follows.

1) We propose UMFCR, a unified network embedding framework that jointly models *communities* and *roles* with two latent embedding spaces, and a fusion decoder that accounts for both within- and across-community interactions.
2) From a Bayesian viewpoint, we design an elaborate fusion pattern to produce the generation process for each link. This process comprehensively considers the dual impact of community and role on the formation of complex networks and realizes the mutual enhancement of these two mechanisms, improving network representation ability.
3) We provide comprehensive experiments over five networks across three tasks, together with ablations and parameter sensitivity, showing competitive or superior performance versus state-of-the-art community- or role-oriented methods.
4) We further add statistical significance tests and discuss complexity, scalability, and limitations to facilitate practical adoption.

II. Related Work

A. Community- or Role-Oriented Network Embedding Methods

Recently, many deep generative models have been proposed and achieved competitive results. The variational graph autoencoder (VGAE) [24] extends the variational auto-encoder (VAE) [32] by using the graph convolution network (GCN) as encoder and inner product as decoder to reconstruct the adjacency matrix. GMMVGAE [33] introduces mixture-of-Gaussians prior to replace the single Gaussian prior to better explaining the data distribution. The deep generative latent feature relational model (DGLFRM) [25] combines stochastic block models and deep generative models to learn the community memberships of nodes, preserving both interpretability and validity.

Beyond classical community detection and role analysis, several recent lines of research have expanded the scope toward overlapping, temporal, and multilayer structures. For example, spectral approaches have been shown to uncover overlapping and hierarchical communities with higher precision by analyzing eigenspaces of network matrices [34]. Dynamic community detection has been advanced by combining network embedding with nonnegative matrix factorization, allowing evolving structures to be identified in time-resolved graphs [35]. In biomedical applications, layer-specific module detection in multilayer cancer networks reveals functionally coherent communities across different molecular interaction layers [36]. Joint representation and clustering frameworks for temporal networks learn feature extraction and clustering simultaneously, enabling large-scale time-evolving graph analysis [37]. Robust methods [38] for incomplete multiview networks leverage joint learning of data recovery and graph contrastive denoising to improve clustering quality under missing views. However, these methods mostly tend to learn community-based node embeddings regardless of roles.

Community-based and role-based node embedding methods have also attracted extensive attention. The former includes DeepWalk [18], GCN [39], SDNE [20], GraphSTONE [19], graphSAGE [40] and so on. Although these approaches model the underlying structure of the network in different ways, they only encode the proximity of nodes in the end. The latter, including RolX [21], DRNE [6], SEGK [26], GraphWave [41], role2vec [42], GAS [23], RESD [22] and RDAA [43], captures the structural identity of nodes. For example, in RolX, GraphWave, and SEGK, matrix factorization techniques are used on structural similarity matrices at different scales. Role2vec proposes a feature-based random walk strategy that treats other nodes with similar connection patterns as context sequences. The rest, including DRNE, GAS, RESD, and RDAA, are based on deep learning, which uses structural features to guide role discovery. However, these methods all ignore the preservation of proximity.

B. Bayesian Generative Models for Communities and Roles

A broad line of probabilistic approaches formalizes community and role structure via Bayesian generative models, where community/role assignments are treated as latent variables. Classical stochastic block models (SBMs) [44] capture structural equivalence through latent classes that govern between-class connectivity, and Bayesian formulations enable principled uncertainty quantification and model selection. Nonparametric variants such as the infinite relational model (IRM) [45] place a Dirichlet-process prior over the number of classes, allowing the model complexity to grow with data. To account for nodes participating in multiple groups, the mixed-membership stochastic blockmodel (MMSB) [46] assigns each node a distribution over communities and samples role/community indicators at the dyad level, yielding state-of-the-art performance for overlapping community discovery and link modeling.

Complementary to class-based blockmodels, latent feature models represent each node by a (potentially infinite) set of binary latent factors drawn from the Indian Buffet Process (IBP), with links generated from feature interactions. This line explicitly treats "roles" as latent factors and has proven effective for link prediction and for capturing heterogeneous role mixtures beyond partition-based communities [47], [48]. Max-margin and deep extensions further enhance predictive performance while retaining Bayesian nonparametric semantics [49].

There are also models that *jointly* explain edges and node attributes under a single Bayesian generative story, thereby integrating community/role inference with node-attribute prediction. The relational topic model (RTM) ties document topics (node attributes) and links via a joint likelihood, improving both link and attribute inference in text networks [50]. The multiplicative attribute graph (MAG) family and the latent multi-group membership graph model (LMMG) connect latent group memberships with observed attributes and edges, enabling community/role discovery together with attribute prediction in large graphs [51], [52].

C. Models for Both Communities and Roles

To date, few works have attempted to unify both community and role memberships of nodes to support subsequent downstream tasks. As far as we know, the approaches most relevant to our work only include Node2vec [28], SPINE [29], REACT [30], and SPaE [31]. Node2vec designs a biased random walk procedure to construct the context sequence of target nodes. The procedure can explore neighborhoods in breadth-first sampling (BFS) as well as depth-first sampling (DFS) fashion, where BFS can closely reflect the structural equivalence of nodes, and DFS can accurately infer communities based on homophily.

Similarly, SPINE defines parameters to encourage sampling strategies that can satisfy learned embeddings preserving local proximity and structural identities simultaneously. REACT and SPaE are both based on matrix factorization, but they apply different methods to unify the community and its role. REACT directly factorizes the adjacency matrix and the RoleSim matrix and integrates their diversity relation as the regularization, while SPaE unifies embedding space and introduces parameters to balance the contribution of proximity and equivalence on the embedding results. Although these methods have shown some gains in various downstream tasks, they usually fail to be both interpretable and effective. They also do not analyze the impact of interactions between different groups from two perspectives on communities and roles in network generation. Therefore, how to integrate community and role to learn better node representations is always an open problem and should be researched.

III. UMFCR

A. Notations and Problem Definition

Given an network $G = (V, E)$, where V denotes N nodes and E denotes their interactions. We leverage an adjacency matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$ to describe such interactions, i.e., having $\mathbf{A}_{ij} = 1$ if $(i, j) \in E$, and $\mathbf{A}_{ij} = 0$ otherwise. $\mathbf{D}_{ii} = \sum_{j=1}^{N} \mathbf{A}_{ij}$ denotes the degree of node i. $\mathbf{X}$ is an $N \times f$ node feature matrix where x_{ij} represents the j-th feature value for the i-th node. The identity matrix will be used in case nodes have no feature available.

In UMFCR, our goal is to automatically learn two sets of D dimensional node embeddings, i.e., $\mathbf{z}^c$ and $\mathbf{z}^r$, for each node by modeling the network formation process. These embeddings should well preserve the community structure and role information of nodes respectively while being capable of serving the network tasks. The detailed description of notions can be seen in Table I. The framework of the proposed UMFCR is shown in Fig. 1.

B. The Generative Process

In UMFCR, we assume that the interactions between nodes are related to both community and role memberships of nodes.

TABLE I
NOTATION SUMMARY USED THROUGHOUT THE ARTICLE

Symbol	Meaning	Symbol	Meaning
$G=(V,E)$	Input graph with node set V and edge set E	$N=\|V\|$	Number of nodes
$\mathbf{A}\in\{0,1\}^{N\times N}$	Adjacency matrix	$\mathbf{D}$	Degree matrix of $\mathbf{A}$
$\tilde{\mathbf{A}}=\mathbf{A}+\mathbf{I}$	Adjacency with self-loops	$\tilde{\mathbf{D}}$	Degree matrix of $\tilde{\mathbf{A}}$
$\mathbf{X}\in\mathbb{R}^{N\times F}$	Node feature matrix (F features)	$\mathbf{I}$	Identity matrix (used as features if absent)
K_1	Number of communities	K_2	Number of roles
$\boldsymbol{\pi}^c\in\Delta^{K_1-1}$	Community membership distribution	$\boldsymbol{\pi}^r\in\Delta^{K_2-1}$	Role membership distribution
$\mathbf{z}_c\in\mathbb{R}^D$	Community-oriented embedding	$\mathbf{z}_r\in\mathbb{R}^D$	Role-oriented embedding
μ_k^c,Σ_k^c	Gaussian mean/covariance for community k	μ_k^r,Σ_k^r	Gaussian mean/covariance for role k
$f_0(\cdot), f_k(\cdot)$	Fusion functions for across/within-community edges	$\sigma(\cdot)$	Sigmoid activation function
$\mathbf{H}^{(\ell)}$	Hidden representations at GCN layer ℓ	$\mathbf{W}^{(\ell)}$	Trainable weight matrix at layer ℓ
$\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}$	Normalized propagation operator	$\mathrm{ReLU}(\cdot)$	Rectified linear activation
$\mathcal{L}_{\mathrm{rec}}$	Reconstruction loss (expected log-likelihood)	$\hat{\mathcal{L}}_{\mathrm{rec}}$	Improved lower bound of $\mathcal{L}_{\mathrm{rec}}$
$\mathcal{L}_{\mathrm{KL}}$	KL divergence regularization	$\mathcal{L}_{\mathrm{deg}}$	Degree-aware role constraint
$\mathcal{L}$	Overall training objective	λ	Weight of reconstruction term
D	Embedding dimension	F	Feature dimension
$p(\cdot), q(\cdot)$	Prior and variational posteriors	ELBO	Evidence lower bound
$\mathbb{E}[\cdot]$	Expectation operator	$\\|\cdot\\|_2$	ℓ_2 norm

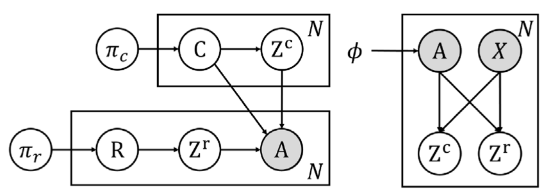

Fig. 1. Graphical representation for our method UMFCR. Left: the generative model. Right: the inference model.

Specifically, support that there are K_1 community clusters and K_2 role clusters, whose prior distribution is controlled by community proportion vector $\boldsymbol{\pi}^c\in\mathbb{R}_+^{K_1}$ and role proportion vector $\boldsymbol{\pi}^r\in\mathbb{R}_+^{K_2}$, where $\sum_{i=1}^{K_1}\boldsymbol{\pi}_i^c=1$ and $\sum_{j=1}^{K_2}\boldsymbol{\pi}_j^r=1$. To generate a link, one can first sample the communities and roles for each node according to the above proportion vectors, then sample the node embeddings from the Gaussian distribution corresponding to sampled clusters, and finally learn the mapping relationship between role-oriented embeddings and role interactions. Its generative process can be described as follows.

1) For each node $i\in V$
   a) Draw a community cluster $c\sim\mathrm{Cat}(\boldsymbol{\pi}^c)$
   b) Draw a role cluster $r\sim\mathrm{Cat}(\boldsymbol{\pi}^r)$
   c) Draw a community-based vector $\mathbf{z}_i^c\sim\mathcal{N}(\boldsymbol{\mu}_c,\boldsymbol{\sigma}_c^2\mathbf{I})$
   d) Draw a role-based vector $\mathbf{z}_i^r\sim\mathcal{N}(\boldsymbol{\mu}_r,\boldsymbol{\sigma}_r^2\mathbf{I})$
2) For each pair of nodes (i,j)
   a) Compute the expectation probability μ_a

$$\mu_a\begin{cases}=\sigma(f_0(\mathbf{z}_i^r)^\top f_0(\mathbf{z}_j^r)+\mathbf{z}_i^{c\top}\mathbf{z}_j^c), & \text{if } c_i\neq c_j\\ =\sigma(f_k(\mathbf{z}_i^r)^\top f_k(\mathbf{z}_j^r)+\mathbf{z}_i^{c\top}\mathbf{z}_j^c), & \text{if } c_i=c_j=k.\end{cases} \tag{1}$$

   b) Sample the value of their interaction

$$a_{ij}\sim\mathrm{Bernoulli}(\mu_a) \tag{2}$$

where $\mathrm{Cat}(\boldsymbol{\pi})$ is the categorical distribution parametrized by $\boldsymbol{\pi}$, $\mathcal{N}(\boldsymbol{\mu}_c,\boldsymbol{\sigma}_c^2\mathbf{I})$ is the Gaussian distribution parametrized by the mean $\boldsymbol{\mu}_c$ and the variance $\boldsymbol{\sigma}_c^2$ related to community c, $\mathbf{I}$ is an identity matrix, $\mathbf{z}_i^c$ and $\mathbf{z}_i^r$ are the community-oriented and role-oriented embedding of node i, $\sigma(\cdot)$ denotes the sigmoid function. $f_k(\cdot)$ is a MLP to learn the interactions between roles within community k, while $f_0(\cdot)$ is used to fit the global interactions of roles between different communities. These MLPs' parameters are denoted as $\boldsymbol{\theta}_k, k\in\{0,1,\ldots,K_1\}$. μ_a indicates the link generation expectation probability, and $\mathrm{Bernoulli}(\cdot)$ is parametrized by μ_a. The elaborate fusion pattern allows us to flexibly model specific role interaction mechanisms for each community. Based on the discussion above, the joint distribution in UMFCR can be described as follows:

$$p(a,\mathbf{z}^c,\mathbf{z}^r,c,r)=p(a|\mathbf{z}^c,\mathbf{z}^r)p(\mathbf{z}^c|c)p(c)p(\mathbf{z}^r|r)p(r) \tag{3}$$

where each term probability can be written as

$$p(c)=\mathrm{Cat}(c|\boldsymbol{\pi}^c) \tag{4}$$
$$p(r)=\mathrm{Cat}(r|\boldsymbol{\pi}^r) \tag{5}$$
$$p(\mathbf{z}^c|c)=\mathcal{N}(\mathbf{z}^c|\boldsymbol{\mu}_c,\boldsymbol{\sigma}_c^2\mathbf{I}) \tag{6}$$
$$p(\mathbf{z}^r|r)=\mathcal{N}(\mathbf{z}^r|\boldsymbol{\mu}_r,\boldsymbol{\sigma}_r^2\mathbf{I}) \tag{7}$$
$$p(a|\mathbf{z}^c,\mathbf{z}^r,c,r)=\mathrm{Bernoulli}(a|\mu_a). \tag{8}$$

C. The Inference Process

In this section, we introduce the inference model $q_{\boldsymbol{\phi}_c}(\mathbf{z}^c|a,\mathbf{x})$ and $q_{\boldsymbol{\phi}_r}(\mathbf{z}^r|a,\mathbf{x})$ to approximate the true distribution $p(\mathbf{z}^c|a,\mathbf{x})$ and $p(\mathbf{z}^r|a,\mathbf{x})$. Let us consider a network $G=(V,E)$, we have an adjacency matrix $\mathbf{A}$, and a node feature matrix $\mathbf{X}$ if available. Similar to the VGAE, we construct a two-layer graph convolutional network (GCN) [39] encoder as the inference models for both the community-oriented and role-oriented embeddings. We take the encoding of community information as an example, that of the roles is similar:

$$[\widetilde{\boldsymbol{\mu}}_c;\log\widetilde{\boldsymbol{\sigma}}_c^2]=g_c(a,\mathbf{x};\boldsymbol{\phi}_c) \tag{9}$$

where the mean $\widetilde{\boldsymbol{\mu}}_c$ and the variance $\widetilde{\boldsymbol{\sigma}}_c^2$ of the approximate posterior are outputs of convolution function $g_c(\cdot)$ with variational parameters $\boldsymbol{\phi}_c$. The propagation rule of convolution is conducted as follows:

$$\mathbf{H}_c^{(l)}=\mathrm{ReLU}(\widetilde{\mathbf{A}}\mathbf{H}_c^{(l-1)}\mathbf{W}_c^{(l)}) \tag{10}$$

where $\mathbf{H}_c^{(l)}$ and $\mathbf{W}_c^{(l)}$ denote the output and trainable weight matrix of the lth layer convolution, $\mathbf{H}_c^{(0)} = \mathbf{X}$, $\widetilde{\mathbf{A}} = \widetilde{\mathbf{D}}^{-1/2}(\mathbf{A}+\mathbf{I})\widetilde{\mathbf{D}}^{-1/2}$, $\widetilde{\mathbf{D}}_{ii} = \sum_{j=1}^{N} \widetilde{\mathbf{A}}_{ij}$. For node embeddings, we utilize a reparameterization trick to ensure that the derivative of $\widetilde{\boldsymbol{\mu}}_c$ and $\widetilde{\boldsymbol{\sigma}}_c^2$ can be computed as

$$\mathbf{z}_i^c = \widetilde{\boldsymbol{\mu}}_{i,c} + \widetilde{\boldsymbol{\sigma}}_{i,c} \odot \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{11}$$

where $\odot$ is the element-wise multiplication.

D. Learning

The learning of UMFCR is to infer its latent variables, i.e., $\mathbf{z}^c, \mathbf{z}^r, \boldsymbol{\theta} = \{\boldsymbol{\pi}^c, \boldsymbol{\pi}^r, \boldsymbol{\mu}_c, \boldsymbol{\mu}_r, \boldsymbol{\sigma}_c^2, \boldsymbol{\sigma}_r^2, \boldsymbol{\theta}_k\}$, and $\boldsymbol{\phi} = \{c, \boldsymbol{\phi}_c, \boldsymbol{\phi}_r\}$, which can be optimized by maximizing the evidence lower bound (ELBO)

$$\begin{aligned}\mathcal{L}_{\text{ELBO}} &= E_{q_{\boldsymbol{\phi}}(\mathbf{z}^c,\mathbf{z}^r,c,r|a,\mathbf{x})}\left[\log \frac{p(a,\mathbf{z}^c,\mathbf{z}^r,c,r)}{q_{\boldsymbol{\phi}}(\mathbf{z}^c,\mathbf{z}^r,c,r \mid a,\mathbf{x})}\right] \\ &= E_{q_{\boldsymbol{\phi}}(\mathbf{z}^c,\mathbf{z}^r,c,r|a,\mathbf{x})}\left[\log p(a \mid \mathbf{z}^c,\mathbf{z}^r,c,r)\right] \\ &\quad - D_{KL}(q_{\boldsymbol{\phi}}(\mathbf{z}^c,\mathbf{z}^r,c,r \mid a,\mathbf{x}) || p(\mathbf{z}^c,\mathbf{z}^r,c,r))\end{aligned} \tag{12}$$

where the first term indicates the reconstruction loss, denoted as $\mathcal{L}_{rec}$, and the second term is regularizer, denoted as $\mathcal{L}_{kl}$, which represents the Kullback–Leibler divergence between approximate posterior and the true prior. On the one hand, the reconstruction loss constrains UMFCR to better preserve the network topology. On the other hand, the regularizer requires the inference model to be closer to the mixture-of-Gaussians prior. Similar to VaDE [53], $q_{\boldsymbol{\phi}}(\mathbf{z}^{\mathbf{c}}, \mathbf{z}^{\mathbf{r}}, c, r|a, \mathbf{x})$ is assumed as a mean-field distribution

$$\begin{aligned}&q_{\boldsymbol{\phi}}(\mathbf{z}^c,\mathbf{z}^r,c,r|a,\mathbf{x}) \\ &= q_{\boldsymbol{\phi}_c}(\mathbf{z}^c|a,\mathbf{x}) q_{\boldsymbol{\phi}_r}(\mathbf{z}^r|a,\mathbf{x}) q(c|a,\mathbf{x}) q(r|a,\mathbf{x}).\end{aligned} \tag{13}$$

Therefore, by substituting the terms in (3)–(13), ideal $\mathcal{L}_{\text{ELBO}}$ and $\mathcal{L}_{rec}$, $\mathcal{L}_{kl}$ can be written as

$$\begin{aligned}\mathcal{L}_{\text{ELBO}} &= E_{q_{\boldsymbol{\phi}}(\mathbf{z}^c,\mathbf{z}^r,c|a,\mathbf{x})}\left[\log \frac{p(a,\mathbf{z}^c,\mathbf{z}^r,c)}{q_{\boldsymbol{\phi}}(\mathbf{z}^c,\mathbf{z}^r,c \mid a,\mathbf{x})}\right] \\ &= E_{q_{\boldsymbol{\phi}}(\mathbf{z}^c,\mathbf{z}^r,c|a,\mathbf{x})}[\log p(a \mid \mathbf{z}^c,\mathbf{z}^r) + \log p(\mathbf{z}^c \mid c) \\ &\quad + \log p(c) + \log p(\mathbf{z}^r \mid r) + \log p(r) - \log q(c \mid a,\mathbf{x}) \\ &\quad - \log q_{\boldsymbol{\phi}_c}(\mathbf{z}^c|a,\mathbf{x}) - \log q_{\boldsymbol{\phi}_r}(\mathbf{z}^r|a,\mathbf{x})]\end{aligned} \tag{14}$$

$$\mathcal{L}_{rec} = \frac{1}{N^2}\sum_{i,j} \mathbf{A}_{ij} \log \mu_a|_{ij} + (1-\mathbf{A}_{ij})\log(1-\mu_a|_{ij}) \tag{15}$$

$$\begin{aligned}\mathcal{L}_{kl} = &-\frac{1}{2}\sum_{c=1}^{K_1}\gamma_c\sum_{d=1}^{D}\left(\log \boldsymbol{\sigma}_c^2|_d + \frac{\widetilde{\boldsymbol{\sigma}}_c^2|_d}{\boldsymbol{\sigma}_c^2|_d} + \frac{(\widetilde{\boldsymbol{\mu}}_c|_d - \boldsymbol{\mu}_c|_d)^2}{\boldsymbol{\sigma}_c^2|_d}\right) \\ &+ \sum_{c=1}^{K_1}\gamma_c \log \frac{\boldsymbol{\pi}_c^c}{\gamma_c} + \frac{1}{2}\sum_{d=1}^{D}(1 + \log \widetilde{\boldsymbol{\sigma}}_c^2|_d) \\ &- \frac{1}{2}\sum_{r=1}^{K_2}\gamma_r\sum_{d=1}^{D}\left(\log \boldsymbol{\sigma}_r^2|_j + \frac{\widetilde{\boldsymbol{\sigma}}_r^2|_d}{\boldsymbol{\sigma}_r^2|_d} + \frac{(\widetilde{\boldsymbol{\mu}}_r|_d - \boldsymbol{\mu}_r|_d)^2}{\boldsymbol{\sigma}_r^2|_d}\right) \\ &+ \sum_{r=1}^{K_2}\gamma_r \log \frac{\boldsymbol{\pi}_r^r}{\gamma_r} + \frac{1}{2}\sum_{d=1}^{D}(1 + \log \widetilde{\boldsymbol{\sigma}}_r^2|_d)\end{aligned} \tag{16}$$

where $\mu_a|_{ij}$ represents the expected interaction probability between node i and j. $*|_d$ is the d-th element of $*$. $\boldsymbol{\pi}_c^c$ is the c-th element of $\boldsymbol{\pi}^c$. $\gamma_c = q(c|a,\mathbf{x}) = p(c|\mathbf{z}^c) \equiv p(c)p(\mathbf{z}^c)/\sum_{c'=1}^{K_1} p(c')p(\mathbf{z}^c \mid c')$, $\gamma_r = q(r|a,\mathbf{x}) = p(r|\mathbf{z}^r) \equiv p(r)p(\mathbf{z}^r)/\sum_{r'=1}^{K_2} p(r')p(\mathbf{z}^r \mid r')$. For details, please refer to VaDE. However, it is risky for this model to degenerate to the trivial solution, i.e., $f_k(\mathbf{z}_i^r)^\top f_k(\mathbf{z}_i^r) \approx 0, k \in \{0,1,\ldots,K_1\}$. To address this problem, we make changes in two places. First, we introduce a degree regularization term denoted as $\mathcal{L}_{deg}$ like DRNE [6] as weak guidance to constrain the learned role-oriented node embeddings to preserve role information

$$\mathcal{L}_{deg} = -\sum_{i=1}^{N}(\log(d(i)+1) - \text{MLP}(\mathbf{z}_i^r))^2 \tag{17}$$

where $d(i)$ denotes degree of node i and $\text{MLP}(\cdot)$ denotes a MLP with ReLU function. Second, we reconstruct an adjacency matrix from the community and role perspectives, respectively, and then calculate the sum of these two losses to replace the ideal reconstruction loss. We mathematically justify its reasonable. Knowing that if $\sigma(\cdot)$ is a sigmoid function, $-\log\sigma(\cdot)$ is a monotonically decreasing convex function. Given $\mu_a^c|_{ij} = \mathbf{z}_i^{c\top}\mathbf{z}_j^c \geq 0$ and $\mu_a^r|_{ij} = f_k(\mathbf{z}_i^r)^\top f_k(\mathbf{z}_j^r) \geq 0$, we have

$$\begin{aligned}-\log \mu_a|_{ij} &= -\log\sigma(\mu_a^c|_{ij} + \mu_a^r|_{ij}) \\ &\leq -\log\sigma\left(\frac{1}{2}\mu_a^c|_{ij} + \frac{1}{2}\mu_a^r|_{ij}\right) \\ &\leq -\frac{1}{2}[\log\sigma(\mu_a^c|_{ij}) + \log\sigma(\mu_a^r|_{ij})]\end{aligned} \tag{18}$$

where we use Jensen's inequality. Based on this, we obtain

$$\begin{aligned}\mathcal{L}_{rec} \geq \hat{\mathcal{L}}_{rec} = &\frac{1}{N^2}\sum_{i,j}\mathbf{A}_{ij}[\log\sigma(\mu_a^c|_{ij}) + \log\sigma(\mu_a^r|_{ij})] \\ &+ (1-\mathbf{A}_{ij})[\log\sigma(1-\mu_a^c|_{ij}) + \log\sigma(1-\mu_a^r|_{ij})].\end{aligned} \tag{19}$$

Thus, we can improve the lower bound of the $\mathcal{L}_{\text{ELBO}}$ by maximizing $\hat{\mathcal{L}}_{rec}$ to indirectly improve the $\mathcal{L}_{\text{ELBO}}$.

To sum up, we train our method UMFCR by jointly maximizing the objective function as follows:

$$\mathcal{L} = \lambda\hat{\mathcal{L}}_{rec} + \mathcal{L}_{kl} + \mathcal{L}_{deg} \tag{20}$$

where λ is the weight of reconstruction loss.

E. Computational Complexity Analysis

UMFCR consists of three main stages: (1) a graph encoder that computes node-level representations via a multilayer GCN backbone; (2) latent sampling and inference that maps encoder outputs to two latent spaces (community-oriented embeddings $\mathbf{z}_c$ and role-oriented embeddings $\mathbf{z}_r$); and (3) a fusion-based decoder that scores edges via a combination of community- and role-based interactions and the associated loss computations. We analyze time costs per training epoch as follows.

A single GCN layer with sparse adjacency requires $O(|E|D)$ multiply-accumulate operations (where $|E|$ is the number of edges and D the embedding dimension). If L layers are used, the encoder cost is $O(L|E|D)$.

TABLE II
BRIEF STATISTICS OF DATASETS WHERE C AND R DENOTE THAT THE NETWORK IS USED FOR COMMUNITY DETECTION AND ROLE DISCOVERY, RESPECTIVELY

Dataset	Nodes	Edges	Feature	Classes	Task
Cora	2708	5278	1433	7	C
Citeseer	3327	4676	3703	6	C
Pubmed	19717	44327	500	3	C
Brazil	131	1074	-	4	R
Europe	399	5993	-	4	R

Mapping encoder outputs to latent parameters (e.g., means and variances for Gaussian embeddings, or logits for membership distributions) is implemented with small MLP heads. These are node-wise dense operations costing $O(ND)$ per head; with H heads (here two main heads for $\mathbf{z}_c, \mathbf{z}_r$) the total cost is $O(HND)$, which is negligible relative to the sparse $|E|D$ encoder when the graph is not extremely sparse.

A naive full-pairwise decoder (computing scores for all $\binom{N}{2}$ node pairs) would cost $O\left(N^2 D\right)$ and be infeasible. In practice, the reconstruction objective is computed using the observed edges and a fixed number S of negative samples per positive edge. Thus the decoder cost per epoch is $O((|E| + S|E|)D) = O(|E|(1+S)D)$. When S is a small constant, this keeps the decoder cost linear in $|E|D$.

Combining encoder, latent heads, and sampled decoder, the per-epoch time is shown as follows:

$$O(L|E|D + HND + S|E|D) \approx O((L+S)|E|D + HND).$$

For typical settings (L small, H small, S small), the dominant term is $O(|E|D)$.

IV. EXPERIMENTS

In this section, we conduct comprehensive experiments over five real-world networks on three downstream tasks including node classification, clustering, and link prediction to evaluate the performance of our proposed UMFCR by comparing with state-of-the-art methods: VGAE [24], GMMVGAE [33], DGLFRM [25], DeepWalk [18], RolX [21], DRNE [6], GraphWave [41], role2vec [42], and node2vec [28].

A. Datasets

We conducted experiments on two types of real-world data sets: (1) citation network for community detection, and (2) air-traffic network for role discovery. The former includes Cora, Citeseer, and Pubmed [8], and the latter includes Brazilian and European air-traffic networks [54]. The statistics are shown in Table II, and the detailed information of these datasets is introduced as follows.

1) *Cora:* A citation network of machine learning articles. It contains $N = 2708$ nodes (articles) and $E = 5429$ undirected edges (citations). Each article is described by a bag-of-words vector of $F = 1433$ binary features, and labeled with one of 7 research topics. We remove multi-edges, symmetrize the citation links, and normalize features row-wise.
2) *Citeseer:* Another citation network with $N = 3327$ nodes and $E = 4732$ undirected edges. Each node has $F = 3703$ binary word features, and is assigned to one of 6 categories.
3) *Pubmed:* A larger biomedical citation network with $N = 19717$ scientific articles and $E = 44338$ undirected links. Each article has a TF-IDF weighted feature vector with $F = 500$ dimensions, and belongs to one of 3 classes. We apply L_2 normalization to the features after TF-IDF transformation. Edges are symmetrized and multiedges are removed.
4) *Brazil Air-Traffic Network:* An air-transportation network where nodes are airports in Brazil and edges represent flight connections. It contains $N = 131$ nodes and $E = 1038$ undirected edges. As no node features are available, we use an identity matrix $\mathbf{I}$ as input features.
5) *Europe Air-Traffic Network:* Similar to the Brazil dataset but on a larger scale. It has $N = 399$ European airports and $E = 5995$ undirected connections. No node attributes are provided, so we use identity features as in the Brazil dataset.

B. Experimental Setting

The baselines with popular techniques range from generative models to network embedding methods. For a fair comparison across baselines, we fix the embedding dimension to $D = 16$, which is the most commonly used setting in the literature. Other hyper-parameters of the baselines are kept as recommended in their original articles or official implementations, which ensures a fair and widely accepted comparison protocol. In UMFCR, we construct the two-layer GCN with the ReLU function as the encoder. For the decoding part, we apply multiple neural networks to learn mapping relationships. For degree regularizer, we use a two-layer MLP with ReLU function. We set the weight of reconstruction loss $\lambda = 6$. We train our model using Adam SGD optimizer for 200 epochs. The model is initialized by pretrain. We use the learned embeddings $\mathbf{z}^c$ and $\mathbf{z}^r$ on citation networks and air-traffic networks, respectively.

All the experiments are conducted on a Linux server (Ubuntu 18.04.6 LTS) equipped with an Intel(R) Xeon(R) CPU E5-2680 v4 @ 2.40GHz. All models are implemented in PyTorch version 1.7.0, Networkx version 2.3.0, pandas version 0.23.4, scikit-learn version 0.20.0, Scipy version 1.5.2 and Python 3.6.9. The link to data and code is https://anonymous.4open.science/r/KDD-07EF.

C. Node Classification

We first conduct experiments on three citation networks, i.e., Cora, Citeseer, and Pubmed, and two air-traffic networks, i.e., Brazilian and European air-traffic networks, to quantitatively evaluate the ability of our model on the node classification task. For each network and method, we randomly select 70% of node representations as input to train a logistic regression classifier, and then test the remaining part. We repeated the experiment 40 times to measure the accuracy with the average value of F1 scores. The results are shown in Table III where OM

TABLE III
NODE CLASSIFICATION PERFORMANCE ON FIVE NETWORKS, WHERE * INDICATES EXPERIMENTS USING INPUT FEATURES IF AVAILABLE

Method	Cora		Citeseer		Pubmed		Brazil		Europe	
	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1
VGAE*	0.5069	0.4300	0.3978	0.3760	0.3400	0.3305	0.3200	0.2400	0.4700	0.2400
GMMVGAE*	0.7143	0.6753	0.6000	0.5600	0.7958	0.7895	0.5163	0.4911	0.4117	0.3835
DGLFRM*	0.7357	0.7168	0.5800	0.5400	0.7658	0.7581	0.4714	0.4518	0.4587	0.4414
DeepWalk	0.7440	0.7303	0.4968	0.4459	0.7911	0.7779	0.3965	0.379	0.3172	0.3096
RolX	0.2232	0.2235	0.2968	0.2742	0.4636	0.4600	0.7427	0.7352	**0.5528**	**0.5416**
DRNE	0.2531	0.2135	0.2037	0.1809	0.3519	0.3291	0.7107	0.6963	0.5511	0.5324
GraphWave	0.2607	0.1915	0.2401	0.1815	0.2985	0.2872	**0.7656**	**0.7557**	0.5221	0.4858
role2vec	0.1927	0.1926	0.1973	0.1855	OM	OM	0.4182	0.4047	0.4079	0.3836
node2vec	0.7368	0.7368	0.5194	0.4852	**0.8083**	**0.7974**	0.3592	0.3422	0.3245	0.3165
UMFCR*	**0.7806**	**0.7538**	**0.6500**	**0.5900**	0.7992	0.7933	0.7612	0.7545	0.5250	0.5022

Note: Bold values indicate the best results.

TABLE IV
PAIRED PERMUTATION TEST P-VALUES COMPARING UMFCR VERSUS EACH BASELINE ON MICRO-F1 AND MACRO-F1

Method	Cora		Citeseer		Pubmed		Brazil		Europe	
	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1
VGAE*	0.0001	0.0001	0.0001	0.0002	0.0001	0.0001	0.0001	0.0001	0.0237	0.0314
GMMVGAE*	0.0105	0.0187	0.0005	0.0011	0.0326	0.0415	0.0001	0.0001	0.0001	0.0001
DGLFRM*	0.0002	0.0003	0.0001	0.0002	0.0005	0.0012	0.0001	0.0001	0.0013	0.0026
DeepWalk	0.0128	0.0159	0.0001	0.0003	0.0403	0.0482	0.0001	0.0001	0.0001	0.0001
RolX	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0178	0.0219	0.0235	0.0184
DRNE	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0013	0.0024	0.0246	0.0197
GraphWave	0.0001	0.0001	0.0001	0.0001	0.0002	0.0004	0.0356	0.0412	0.6097	0.0487
role2vec	0.0001	0.0001	0.0001	0.0001	–	–	0.0001	0.0001	0.0001	0.0001
node2vec	0.0007	0.0014	0.0002	0.0006	0.0327	0.0389	0.0001	0.0001	0.0001	0.0001

means memory limit exceeded. We highlight the best performing results in bold. It is worth noting that VGAE, GMMVGAE, DGLFRM, and UMFCR leverage node features as input when learning low-dimensional representations on three citation networks. We also performed paired permutation tests comparing UMFCR against each baseline (as shown in Table IV). Most of UMFCR's gains are statistically significant ($p \ll 0.05$). The only clear nonsignificant comparison is UMFCR versus GraphWave on the European network ($p \approx 0.61$), where the Micro-F1 score is similar.

From the results, we have the following observations. First, our model outperforms all baselines on the citation networks, except that UMFCR is slightly lower than node2vec on Pubmed. It demonstrates the effectiveness of our model in preserving community information. Intuitively, since community-oriented NE methods or deep generative models are designed to capture the proximity relationships between nodes, their learned node representations can naturally provide beneficial information for community-guided classification tasks. However, methods like RolX, DRNE, GraphWave, and role2vec cannot deal well with such networks where node labels are highly correlated with their neighbors, because role-oriented methods only encode the structural similarity of nodes and ignore the proximity of nodes. GMMVGAE performs better than VGAE because the Mixture-of-Gaussians prior can better explain more complex network data. Second, it is clear that these methods show completely opposite trends in two air-traffic networks than citation networks. However, it is worth noting that while the performance of our model is not optimal, it is still relatively prominent among all baselines and significantly outperforms community-oriented methods, which validates that our model has a certain gain in role discovery tasks. Third, compared with the node2vec, which claims that enables to maintenance of proximity and structural similarity at the same time, our method UMFCR shows stronger advantages.

Furthermore, considering the influence of the training set proportion on the experimental results, we adjust it from 10% to 90%, and then repeat the above experiments. The results can be seen in Fig. 2 where the solid line indicates the experiment without using the node features and the dashed line indicates the experiment using the node features. Intuitively, the classification performance of various methods shows a slow growth trend as the proportion of the training set increases, except VGAE. It is clear that if UMFCR uses node features, the results are optimal compared with other baselines. However, when UMFCR does not leverage features, its performance degrades. This is because node features usually contain richer information.

D. Clustering

Clustering is a common unsupervised task that aims to divide nodes into distinct groups. In this experiment, based on the learned representations, we utilize the K-means algorithm to obtain the predicted labels of nodes and calculate the normalized mutual information (NMI) and adjusted rand index (ARI) as evaluation metrics. Similarly, to make a fair comparison,

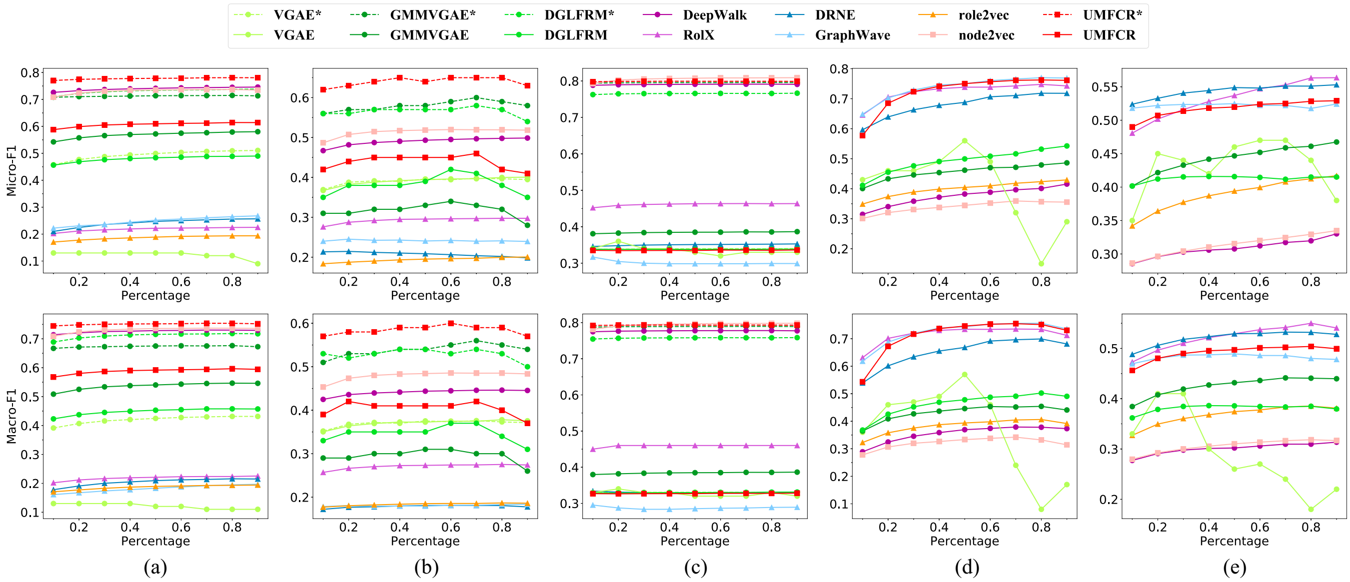

Fig. 2. Node classification performance reported is the average of micro-F1 and macro-F1 with different ratio of training set. (a) Cora network. (b) Citeseer network. (c) Pubmed network. (d) Brazillian air-traffic network. (e) European air-traffic network.

TABLE V
NODE CLUSTERING PERFORMANCE USING NMI AND ARI SCORES, WHERE * INDICATES EXPERIMENTS USING INPUT FEATURES IF AVAILABLE

Method	Cora		Citeseer		Pubmed		Brazil		Europe	
	NMI	ARI	NMI	ARI	NMI	ARI	NMI	ARI	NMI	ARI
VGAE*	0.3067 ± 0.0019	0.2914 ± 0.0018	0.2403 ± 0.0278	0.2283 ± 0.0264	0.0001 ± 0.0001	0.0001 ± 0.0001	0.0214 ± 0.0100	0.0203 ± 0.0095	0.0090 ± 0.0034	0.0086 ± 0.0032
GMMVGAE*	0.5127 ± 0.0048	0.4871 ± 0.0046	0.2753 ± 0.0007	0.2615 ± 0.0007	0.2587 ± 0.0000	0.2328 ± 0.0000	0.2980 ± 0.0427	0.2831 ± 0.0406	0.1500 ± 0.0123	0.1425 ± 0.0117
DGLFRM*	0.4204 ± 0.0015	0.3994 ± 0.0014	0.2445 ± 0.0044	0.2323 ± 0.0042	0.2639 ± 0.0002	0.2375 ± 0.0002	0.2516 ± 0.0140	0.2390 ± 0.0133	0.1420 ± 0.0022	0.1350 ± 0.0021
DeepWalk	0.3452 ± 0.0459	0.3280 ± 0.0436	0.1073 ± 0.0293	0.1020 ± 0.0278	0.2700 ± 0.0002	0.2430 ± 0.0002	0.1168 ± 0.0248	0.1110 ± 0.0236	0.0525 ± 0.0035	0.0499 ± 0.0033
RolX	0.0367 ± 0.0043	0.0349 ± 0.0041	0.0507 ± 0.0012	0.0482 ± 0.0011	0.0115 ± 0.0001	0.0104 ± 0.0001	**0.5443** ± 0.0295	**0.5161** ± 0.0280	0.1017 ± 0.0333	0.0966 ± 0.0316
DRNE	0.0197 ± 0.0015	0.0187 ± 0.0014	0.0231 ± 0.0003	0.0220 ± 0.0003	0.0016 ± 0.0000	0.0014 ± 0.0000	0.4660 ± 0.0114	0.4427 ± 0.0108	**0.3466** ± 0.0007	**0.3293** ± 0.0007
GraphWave	0.0427 ± 0.0000	0.0406 ± 0.0000	0.0422 ± 0.0020	0.0401 ± 0.0019	0.0005 ± 0.0000	0.0005 ± 0.0000	0.5010 ± 0.0146	0.4759 ± 0.0139	0.3214 ± 0.0041	0.3053 ± 0.0039
role2vec	0.0123 ± 0.0018	0.0117 ± 0.0017	0.0084 ± 0.0005	0.0080 ± 0.0005	OM	OM	0.1392 ± 0.0255	0.1322 ± 0.0242	0.0733 ± 0.0089	0.0696 ± 0.0084
node2vec	0.4499 ± 0.0154	0.4274 ± 0.0146	0.2205 ± 0.0152	0.2095 ± 0.0144	**0.2885** ± 0.0002	**0.2597** ± 0.0002	0.0763 ± 0.0083	0.0725 ± 0.0079	0.0438 ± 0.0010	0.0416 ± 0.0010
UMFCR*	**0.5438** ± 0.0004	**0.5166** ± 0.0004	**0.2895** ± 0.0000	**0.2750** ± 0.0000	0.2710 ± 0.0009	0.2439 ± 0.0008	0.4951 ± 0.0128	0.4703 ± 0.0122	0.2918 ± 0.0038	0.2772 ± 0.0036

Note: Bold values indicate the best results.

we repeat the experiment 40 times. The mean and variance of scores are reported in Table V where the best scores are in bold.

As might be expected, our method UMFCR still has good performance on the clustering task, which is consistent with the node classification task. For the other baselines, GMMVGAE is the most competitive on Cora and Citeseer, but its clustering results on Pubmed are mediocre, and not satisfactory on air-traffic networks. On the contrary, although DeepWalk and node2vec perform poorly on Cora and Citeseer, they are comparable to UMFCR on Pubmed, especially node2vec. However, none of them can handle the air-traffic networks. While DGLFRM unifies two directions of stochastic blockmodels and graph convolutional networks to retain excellent predictive performance and interpretability, it is still inferior to GMMVGAE in practical application. The clustering results of the two air-traffic networks are ranked differently from the classification results, with RolX and DRNE performing the best on the Brazilian and European air traffic networks, respectively. However, we can observe that the NMI value of RolX on the European network is only 0.1017, and the clustering ability of DRNE on the Brazilian network is also not very significant. Even though GraphWave performs well on both air-traffic networks, unsurprisingly, their learned node representations barely provide available information for node clustering on three citation networks. In addition, by comparing the values of NMI, it is clear that node2vec does not show encouraging results in node clustering tasks. In contrast, our model UMFCR can automatically learn two sets of low dimensional representations for each node without introducing any prior knowledge and performs well on all the networks, which indicates that it cannot only be effectively used for community detection tasks but also complete role discovery tasks. It has stronger generalizability across different networks. We attribute such success to the design of the integration model between communities and roles in UMFCR, which realizes the mutual reinforcement between these two types of tasks and improves the ability of network representation.

E. Link Prediction

In this part, we conduct a link prediction task to validate the reasonable model design. Specifically, for each network, we randomly sample 85% of the links as our training data to learn node embeddings. The validation set and test set contain 5% and 10% of the remaining links, respectively. The validation set is used for optimization of hyperparameters and the test set is used to measure the ability of the model in link prediction. We

TABLE VI
LINK PREDICTION PERFORMANCE OF FOUR DEEP GENERATIVE MODELS USING AUC, AP, AND MICRO-F1 AS EVALUATION METRICS

Method	Cora			Citeseer			Pubmed			Brazil			Europe		
	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1
VGAE*	0.9255	0.9326	0.8653	0.9128	0.9128	0.8551	0.9506	0.9515	0.9052	0.8295	0.8379	0.7907	0.8459	0.8545	0.8058
GMMVGAE*	0.8433	0.8433	0.7651	0.6979	0.6979	0.6354	0.9566	0.9578	0.9102	0.8588	0.8919	0.8204	0.6723	0.5781	0.6007
DGLFRM*	**0.9400**	**0.9496**	0.8803	0.9179	0.9179	0.8651	**0.9582**	**0.9618**	0.9254	**0.9061**	**0.9239**	**0.8703**	0.8747	0.8807	0.8351
UMFCR*	0.9319	0.9419	**0.9004**	**0.9354**	**0.9354**	**0.8807**	0.9526	0.9516	**0.9403**	0.9061	0.8840	0.8250	**0.8774**	**0.8845**	**0.8509**

Note: Bold values indicate the best results.

TABLE VII
PAIRED PERMUTATION TEST p-VALUES COMPARING UMFCR WITH BASELINE MODELS ON THE LINK PREDICTION EXPERIMENT

Method	Cora			Citeseer			Pubmed			Brazil			Europe		
	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1
VGAE*	0.0120	0.0031	0.0002	0.0018	0.0022	0.0010	0.3316	0.4876	0.0003	0.0001	0.0007	0.0017	0.0009	0.0012	0.0001
GMMVGAE*	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.2124	0.0978	0.0029	0.0003	0.1895	0.3822	0.0001	0.0001	0.0001
DGLFRM*	0.0746	0.0853	0.0217	0.0064	0.0069	0.0389	0.1451	0.0380	0.0458	0.4995	0.0011	0.0002	0.3480	0.2792	0.0307

compare the area under the ROC curve (AUC), average precision (AP), and micro-F1 scores corresponding to the other three deep generative models on five datasets. The average results of 40 times are reported in Table VI, where higher values indicate better performance of the model on the link prediction task. Using paired permutation tests under identical experimental settings, we compare UMFCR against each baseline method and report the corresponding p-values in Table VII. Among all methods, we can observe an interesting phenomenon that scores of GMMVGAE are much worse than other baselines, even lower than the results of VGAE, which is completely different from what we see on the node classification task and clustering task. The results of VGAE on the citation network are similar to DGLFRM and UMFCR, but show a clear disadvantage on two air-traffic networks. DGLFRM and UMFCR perform best, which is sufficient to illustrate that it is reasonable and effective for us to produce the generation process for each link from the perspective of community and role membership distributions and the interactions between them.

Furthermore, in order to eliminate the influence of different training set proportions, we have carried out richer experiments. First, we vary the proportion of training set from 50% to 90% to evaluate the performance of these approaches. Second, we report the experimental results with and without using node features. The results are shown in Fig. 3. It is worth noting that we do not report the performance on Pubmed without using node features. It is because their performances are extremely poor. It is clear that with the increasing of the training ratio, the results of each method are improved to different degrees. We can see that under the same conditions, UMFCR has the best performance on Cora and Citeseer, but slightly falls short on Pubmed. For both air-traffic networks, UMFCR has a clear upward trend compared with other methods. Overall, these performances are consistent with the conclusions above.

F. Parameter Sensitivity

We evaluate the affect performance of the embedding dimension D and the weight of reconstruction loss λ. The results are shown in Fig. 4. For the embedding dimension, we vary D from 2 to 64. We can see that dimension settings have different effects on various types of networks and downstream tasks. On Cora network, Micro-F1 and NMI scores first decrease and then increase with increasing dimensionality, because it allows node embeddings to cover more useful information. When the dimension is equal to 16, it reaches the peak and tends to be stable on node classification but shows a decreasing trend on the clustering task. The reason for the decline may be that the learned node representations contain a lot of other information that is not conducive to clustering. On the European air-traffic network, UMFCR has similar trends in these two tasks. However, for the link prediction, UMFCR shows different trends. For example, it decreases sharply after reaching the peak on the Cora network, while it maintains an upward trend on the European air-traffic network. Based on these conclusion, we set the dimension to 16 in our experiments.

For the weight of the reconstruction loss, we investigate how the contribution of $\hat{\mathcal{L}}_{rec}$ affects model performance by varying λ from 2 to 10. Unlike the effect of the embedding dimension, We can clearly observe that the performance first increases and then gradually flattens out or decreases slightly across different networks and downstream tasks, except for the link prediction experiments on the Cora network. The increase in value of the evaluation indicator demonstrates that the reconstruction loss term is beneficial to constrain the node embedding to better support the network analysis task. However, when the coefficient is too large, it will cause the model to ignore other constraints, resulting in performance degradation.

G. Efficiency Analysis

In this section, we compare the time for generating embeddings with baseline methods to analyze the applicability on large-scale networks. The results can be seen in Fig. 5. We find that some shadow methods (DeepWalk and node2vec) cost lower time on small networks, but perform poorer on larger datasets. Compared with other deep-learning methods, there's no significant gap between our model and them, indicating that

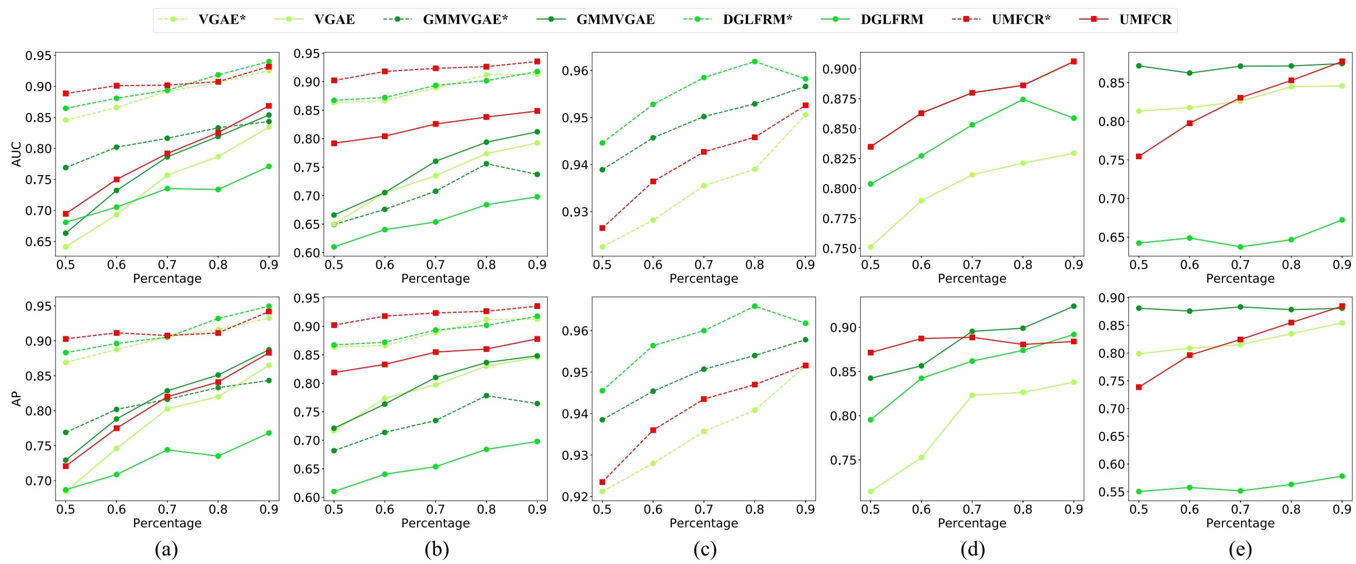

Fig. 3. Link prediction performance of four deep generative methods on three citation networks and two air-traffic networks. (a) Cora network. (b) Citeseer network. (c) Pubmed network. (d) Brazillian air-traffic network. (e) European air-traffic network.

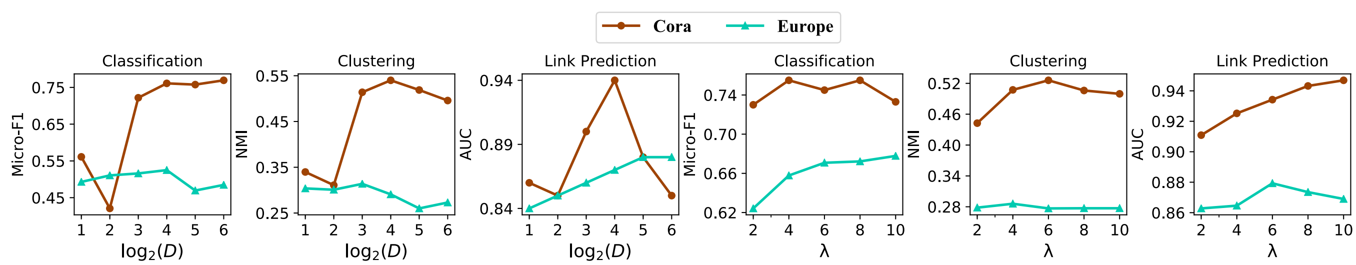

Fig. 4. Parameter sensitivity performance with different embedding dimension D and the weight of construction loss λ.

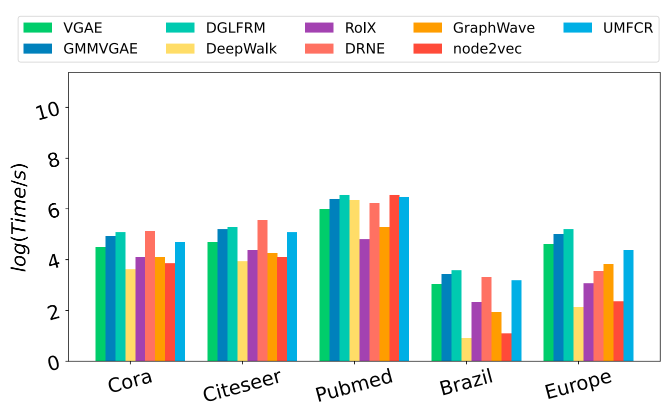

Fig. 5. Time for generating node representations on different networks.

our model can also be applied for large-scale networks with proper computational complexity.

H. Ablation Study

To further explore the effectiveness of each component in UMFCR, we conduct ablation experiments. Specifically, we construct two variants of UMFCR, i.e., UMFCR-F and UMFCR-X, which are described in detail as follows.

1) UMFCR-F. As discussed in Section III, we modify the ideal reconstruction loss by reconstructing an adjacency matrix using community-based embeddings and role-based embeddings, respectively. Here, we provide the original version UMFCR-F to validate the effectiveness of such correction, which trains the model with an ideal reconstruction loss.
2) UMFCR-X. The motivation and contribution of this article is to integrate communities and roles to better model the underlying structure of complex networks. To illustrate the design of the generation process is useful, we provide another variant UMFCR-X, which does not use the fusion mechanism.

Based on the above variants, we conduct comprehensive experiments on citation networks and air-traffic networks. The results are shown in Table VIII. Intuitively, compared to UMFCR-F and UMFCR-X, UMFCR achieves great improvements on all three network tasks, especially on the unsupervised clustering task. These results fully show that the loss function term and fusion pattern of UMFCR are reasonable and effective. More importantly, the experimental results also favorably support our original hypothesis that the generation of the network is related to the community and roles of nodes. We also find that UMFCR-F generally outperforms UMFCR-X with only slightly lower

TABLE VIII
RESULTS OF ABLATION STUDY

Dataset	Metric	UMFCR	UMFCR-F	UMFCR-X
Cora	Micro-F1	**0.7806**	0.7552	0.4887
	NMI	**0.5430**	0.2277	0.3564
	AUC	**0.9400**	0.8900	0.7800
Citeseer	Micro-F1	**0.6500**	0.5666	0.5616
	NMI	**0.2895**	0.0846	0.2554
	AUC	**0.9345**	0.8614	0.7897
Pubmed	Micro-F1	**0.7991**	0.6527	0.7865
	NMI	**0.2710**	0.1845	0.2517
	AUC	**0.9526**	0.9516	0.9501
Brazil	Micro-F1	**0.7625**	0.6544	0.6306
	NMI	**0.4951**	0.4862	0.4857
	AUC	**0.9061**	0.8253	0.4734
Europe	Micro-F1	0.5250	**0.5325**	0.5146
	NMI	0.2918	**0.2971**	0.2912
	AUC	**0.8800**	0.8400	0.7300

Note: Bold values indicate the best results.

scores in a few cases. On the one hand, this result reiterates the effectiveness of fusing concepts of community and role for modeling complex networks. On the other hand, it also indicates that the contribution of the fusion mechanism is much greater than the design of the objective function.

V. CONCLUSION

The fusion of community detection and role discovery is always an open question. We propose a unified deep generative framework to automatically generate two sets of community-oriented and role-oriented low-dimensional representations for each node. We combine variational graph auto-encoder and GMM to infer the community and role membership distributions of nodes and develop an elaborate fusion pattern that enables the interactions among them, thereby simulating the complex network generation process. We conduct comprehensive experiments including node classification, clustering, and link prediction on several real-world networks. The promising experimental results demonstrate the superiority of our model.

While UMFCR demonstrates strong performance and effectively integrates community and role representations, several limitations remain to be studied. Our model focuses on static homogeneous graphs, and extensions to dynamic or heterogeneous networks are not straightforward. Although the asymptotic complexity is comparable to other GCN-based models, the additional inference process and fusion decoder may hinder scalability on very large graphs. Besides, generalization across domains and networks is still an open challenge, and future work should explore transfer learning or meta-learning strategies to enhance robustness. We believe that these limitations will inspire further research and lead to more comprehensive models that combine structural roles, community semantics, and scalability in real-world network applications.

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Pengfei Jiao (Member, IEEE) received the Ph.D. degree in computer science from Tianjin University, Tianjin, China, in 2018.

From 2018 to 2021, he was a Lecturer with the Center of Biosafety Research and Strategy, Tianjin University. He is currently a Professor with the School of Cyberspace, Hangzhou Dianzi University, Hangzhou, China. His research interests include complex network analysis and its applications.

Wang Zhang received the bachelor's degree in 2018, and the master's degree in 2022, both from Tianjin University, Tianjin, China, where he is currently working toward the Ph.D. degree with the College of Intelligence and Computing, all in computer science and technology.

His research interests include complex network analysis and network embedding.

Xuan Guo is currently working toward the Ph.D. degree in computer science and technology with the College of Intelligence and Computing, Tianjin University, Tianjin, China.

His research interests include complex network analysis, role discovery, network representation learning, and percolation model.

Huan Liu received the master's degree in computer science and technology in 2024 from the Computer School, Hangzhou Dianzi University, Hangzhou, China, where he is currently working toward the Ph.D. degree in computer science and technology with the School of Cyberspace.

His research interests include complex network analysis and network embedding.

Yanxian Bi received the Ph.D. degree in computer science and technology from Beihang University, Beijing, China, in 2017.

He is currently a Senior Engineer with China Academy of Electronic and Information Technology, CETC Academy of Electronics and Information Technology Group Company Ltd., Beijing, China. From 2015 to 2017, he studied in United Kingdom of Great Britain and Northern Ireland as a Visiting Scholar funded by China Scholar Council (CSC). His research interest includes artificial intelligence.

Yefei Zhang received the Ph.D. degree in electronic science and technology from Hangzhou Dianzi University, Hangzhou, China, in 2023.

She is currently an Associate Professor with the School of Cyberspace, Hangzhou Dianzi University. She has authored or coauthored over 20 papers. Her research interests include medical artificial intelligence and its research and application in fields such as information security and medical health.

Zhidong Zhao received the B.S. and M.S. degrees in mechanical engineering from Nanjing University of Science and Technology, Nanjing, China, in 1998 and 2001, respectively, and the Ph.D. degree in biomedical engineering from Zhejiang University, Hangzhou, China, in 2004.

He is currently a Full Professor with Hangzhou Dianzi University, China. His research interests include biomedical signal processing, wireless sensor networks, biometrics, and machine learning.